Quality Control Standards

QC-STD-200 | Revision C | Effective 2026-02-10

Field	Value
Facility	Northfield Components Inc. - Plant 2 (Riverside)
Document owner	Quality Assurance (QA) Manager
Approved by	QA Manager; Plant Manager
Applies to	Operators, inspectors, floor supervisors and QA staff at Plant 2
Review cycle	Annually, or when a customer requirement or drawing changes

Note: FICTIONAL SAMPLE DOCUMENT - for software development and testing only. Names, equipment, part numbers and values are invented.

1. Purpose and Scope

This standard defines product specifications, inspection frequencies, sampling plans, measurement requirements and the handling of nonconforming product for parts made at Plant 2: **A-102 Drive Shaft** (Line 1), **B-220 Mounting Bracket** (Line 2) and **C-310 Pump Housing** (Line 1). The released customer drawing always governs; if this standard and the drawing disagree, stop and contact QA.

NOTE: Handle machined and stamped parts with cut-resistant gloves (ANSI A2 or higher) as required by EHS-SOP-001 Section 4.2.

2. Definitions

Term	Meaning
CTQ	Critical-to-quality characteristic. Marked with a diamond on the drawing; requires SPC and 100% traceable records.
FAI	First-article inspection: full check of the first part(s) after a setup, tooling change or start of shift.
AQL	Acceptable quality limit used to choose a sampling plan.
NCR	Nonconformance report, recorded in the QMS.
MRB	Material Review Board: QA Manager, Production Manager and Product Engineer. Only the MRB may disposition nonconforming product.
Cpk	Process capability index: how well a stable process fits within the tolerance, accounting for centering.
TIR	Total indicator reading (runout measured with a dial indicator).

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3. Part Specifications

3.1 A-102 Drive Shaft (Line 1, CNC-201/202)

Material: 4140 steel bar, pre-hardened. Customer drawing A-102 Rev F.

Characteristic	Specification	CTQ	Gauge
Bearing journal diameter (2 places)	25.00 mm ±0.02 mm (24.98-25.02)	Yes	Micrometer 0-25 / 25-50 mm, 0.001 mm resolution
Overall length	150.0 mm ±0.1 mm	No	Height gauge or caliper
Keyway width	8.00 mm +0.03 / -0.00 mm	Yes	Go/no-go keyway gauge
Keyway depth	4.0 mm ±0.1 mm	No	Depth micrometer
Journal runout	0.02 mm TIR maximum	Yes	Dial indicator on V-blocks
Journal surface finish	Ra 0.8 µm maximum	No	Surface roughness tester
Hardness	28-32 HRC	No	Rockwell hardness tester
Chamfers	0.5 mm x 45° ±0.2 mm, no burrs	No	Visual and chamfer gauge

3.2 B-220 Mounting Bracket (Line 2, HP-400 and RW-510)

Material: 4 mm hot-rolled steel, pressed on HP-400 and welded on RW-510. Customer drawing B-220 Rev C.

Characteristic	Specification	CTQ	Gauge
Mounting hole spacing	80.0 mm ±0.2 mm	Yes	Check fixture CF-220 or caliper
Mounting hole diameter	10.5 mm +0.2 / -0.0 mm	No	Pin gauges 10.5 (go) / 10.7 (no-go)
Bend angle	90° ±0.5°	Yes	Digital protractor
Flatness of mounting face	0.3 mm maximum	No	Feeler gauge on surface plate
Weld length (2 fillet welds)	25 mm minimum each	Yes	Weld gauge / ruler
Weld size	4 mm fillet leg minimum	No	Fillet weld gauge
Weld quality	No cracks, no incomplete fusion; porosity max 1 mm pore, max 2 pores per weld	Yes	Visual (Section 9)

3.3 C-310 Pump Housing (Line 1, CNC-203)

Material: 6061-T6 aluminum casting, machined. Customer drawing C-310 Rev B.

Characteristic	Specification	CTQ	Gauge
Main bore diameter	52.000 mm ±0.015 mm	Yes	Bore gauge, set with master ring
Sealing face flatness	0.05 mm maximum	Yes	CMM
Mounting threads (4x)	M8 x 1.25-6H	No	Thread plug gauge go/no-go
Bore surface finish	Ra 1.6 µm maximum	No	Surface roughness tester
Leak test	Air at 6 bar (87 psi) for 30 s; max leak 2 cc/min	Yes	Leak tester LT-1, 100% of parts

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4. Inspection Plan

4.1 First-Article Inspection (FAI)

An FAI of **all characteristics** in Section 3 is required on the first part after any of the following, and before production continues:

- Start of each shift.
- Machine setup or program change.
- Tool insert change affecting a CTQ, and any die change on HP-400 (MM-100 Section 5.4).
- Machine repair or crash, or return to service after maintenance that touched the machine axes, die or fixtures.
- New material lot.

The operator measures the part and the floor supervisor or an inspector verifies and signs the FAI record. Production may not continue until the FAI passes.

4.2 In-Process Inspection Frequency

Part	Characteristic	Frequency	Recorded by
A-102	Journal diameter (both)	Every 25 parts (SPC subgroup of 5 consecutive parts)	Operator, SPC chart
A-102	Runout and keyway width	Every 50 parts	Operator
A-102	Hardness	Once per material lot	QC inspector
B-220	Bend angle and hole spacing	Every 30 parts	Operator
B-220	Weld visual	100% of parts	Operator
B-220	Destructive weld test (cut and etch)	1 part per shift	QC inspector
C-310	Bore diameter	Every 20 parts (SPC subgroup of 5)	Operator, SPC chart
C-310	Leak test	100% of parts	Leak tester LT-1 (automatic record)
C-310	Sealing face flatness (CMM)	2 parts per shift	QC inspector

4.3 Final Inspection

Before packing, each lot is sampled per Section 5 for visual defects and the CTQ characteristics. Packing on Line 3 may not seal a container until the lot has a green QC RELEASED label.

5. Sampling Plan for Lot Acceptance

Final inspection and incoming inspection of purchased material use the plant's zero-acceptance (c=0) plan below. **Any** defect found in the sample rejects the lot: the lot is placed on hold and an NCR is raised (Section 8).

Lot size	Sample size (non-CTQ characteristics)	Sample size (CTQ characteristics)	Accept / reject
2-50	8 (or all if fewer)	All parts	Accept 0 / reject 1

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Lot size	Sample size (non-CTQ characteristics)	Sample size (CTQ characteristics)	Accept / reject
51-150	13	20	Accept 0 / reject 1
151-500	29	47	Accept 0 / reject 1
501-1,200	34	53	Accept 0 / reject 1
1,201-3,200	42	60	Accept 0 / reject 1
3,201-10,000	50	74	Accept 0 / reject 1

Characteristics inspected 100% (C-310 leak test, B-220 weld visual) are not re-sampled; the automatic or operator records are reviewed instead.

6. Measuring Equipment and Calibration

Equipment	Resolution	Calibration interval	Daily check
Digital calipers	0.01 mm	6 months	Zero check; check against 25 mm gauge block
Outside micrometers	0.001 mm	6 months	Zero check with standard
Bore gauge	0.001 mm	6 months	Set with 52.000 mm master ring at start of shift
Go/no-go gauges and pin gauges	-	12 months	Visual for wear and damage
Digital protractor	0.1°	12 months	Check on 90° square
CMM	0.001 mm	12 months (external)	Probe qualification on reference sphere
Rockwell hardness tester	0.1 HRC	12 months (external)	Test block within ±1.0 HRC of its value
Leak tester LT-1	0.1 cc/min	12 months	Master leak part must fail; master good part must pass

- Every gauge carries a green calibration sticker with its ID and due date. **Never use a gauge that is past due, damaged or unlabeled.**
- If a gauge is found out of calibration, quarantine it and notify QA. QA evaluates all product measured with that gauge since its last good calibration.
- Gauge R&R; for CTQ gauges: under 10% of tolerance is acceptable; 10-30% is conditionally acceptable with QA approval; over 30% is not acceptable.
- Personal gauges are not allowed for product acceptance.

7. Statistical Process Control (SPC)

CTQ characteristics measured in-process (A-102 journal diameter, C-310 bore diameter) are plotted on X-bar and R charts using subgroups of 5 consecutive parts. Control limits are recalculated by QA after any process change or every 3 months.

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7.1 Capability Requirements

- Cpk must be at least **1.33** for every CTQ characteristic in ongoing production.
- New or changed processes must show Cpk of at least 1.67 over 30 subgroups before release.
- If Cpk falls below 1.33, QA adds 100% inspection of that characteristic until the process is corrected.

7.2 Out-of-Control Rules

The process is out of control if any of the following occur on the chart:

1. One point outside the 3-sigma control limits.
2. Seven consecutive points on the same side of the center line.
3. Six consecutive points steadily increasing or decreasing.
4. Two of three consecutive points beyond the 2-sigma line on the same side.

7.3 Reaction Plan

1. Stop the process and notify the floor supervisor.
2. Hold all parts made since the last in-control subgroup and check them 100% for the affected characteristic.
3. Find and correct the cause (for example tool wear, missed spindle warm-up per MM-100 Section 6.3, or offset error), and note it on the chart.
4. Run an FAI (Section 4.1) before restarting production.

A point outside the control limits is not automatically a nonconforming part. A part is nonconforming only when it is outside the **specification** limits in Section 3; then Section 8 also applies.

8. Nonconforming Product

8.1 Identify and Segregate

Immediately attach a red **HOLD** tag to any nonconforming or suspect part or container and move it to the red-marked quarantine cage at the line. Nonconforming parts must never be left at the workstation or mixed with good parts.

8.2 Report

The floor supervisor or inspector opens an NCR in the QMS **within 4 hours** of discovery, recording the part number, lot, quantity, characteristic, measured values and how the problem was found.

8.3 Contain

Contain all product made since the last good inspection, including parts in work-in-process, finished goods and the shipping dock. If any suspect product may have shipped, QA must notify the customer **within 24 hours**.

8.4 Disposition

Only the MRB may disposition nonconforming product; **floor supervisors and operators may not use, rework or scrap held product on their own**. The MRB meets daily at 10:00. Dispositions are:

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Disposition	When used	Requirements
Use as is	Deviation does not affect form, fit or function	Engineering approval; customer approval for any CTQ deviation
Rework	Part can be brought into specification	Approved written rework instruction; 100% re-inspection after rework
Scrap	Part cannot be made conforming	Mark with red paint, place in scrap bin, record quantity in QMS
Return to supplier	Purchased material defect	Supplier corrective action request (SCAR) issued by QA

8.5 Corrective Action

Repeat problems (same defect 3 times in 30 days), customer complaints and any escape to a customer require a formal corrective action using 8D, with containment within 24 hours and root cause within 10 working days.

9. Visual and Workmanship Standards

Inspect under at least 1,000 lux lighting at a viewing distance of about 450 mm (18 in), spending about 10 seconds per part surface.

Class	Examples	Allowed
Critical	Cracks; missing or incomplete welds; wrong part or material; failed leak test	None. Lot on hold, NCR required.
Major	Burrs over 0.2 mm on functional edges or bores; weld porosity over 1 mm; damaged threads; dents on sealing faces	None in the sample (c=0).
Minor	Light scratches under 0.1 mm deep on non-functional surfaces; minor discoloration; light weld spatter away from mounting faces	Acceptable. Note trends on the shift log.

10. Records and Retention

Record	Retention
FAI, in-process and final inspection records	7 years
SPC charts	7 years
NCRs, MRB decisions, corrective actions	10 years
Calibration records	Life of the gauge plus 3 years
Leak test data (C-310)	10 years (customer requirement)

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11. Escalation

Situation	Escalate to	Timing
Any CTQ out of specification	Floor supervisor and QA inspector	Immediately
Process out of control (Section 7.2)	Floor supervisor	Immediately
Suspect product may have shipped	QA Manager	Within 1 hour
Gauge found out of calibration	QA	Before end of shift
Quality issue that is also a safety issue (for example a cracked weld on a guard bracket)	QA Manager and EHS Manager	Immediately

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